CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

NAAC GRADE A+
Accredited University

Experiment 2

Student Name Ayush | **UID** 22bcs15510
Branch: CSE | **Section/Group** 608-A
Semester: 6th | **Date of Performance:** 21/01/2025
Subject Name: Java Lab | **Subject Code:** 22CSH-359

1. **Aim :** The aim of this project is to design and implement a simple inventory control system for a small video rental store. Define least two classes: a class Video to model a video and a class VideoStore to model the actual store.

   Assume that an object of class Video has the following attributes:

   1. A title;
   2. a flag to say whether it is checked out or not;
   3. An average user rating.

   Add instance variables for each of these attributes to the Video class.

   In addition, you will need to add methods corresponding to the following:

   1. being checked out;
   2. being returned;
   3. receiving a rating.

   The VideoStore class will contain at least an instance variable that references an array of videos (say of length 10). The VideoStore will contain the following methods:

   1. addVideo(String): add a new video (by title) to the inventory;
   2. checkOut(String): check out a video (by title);
   3. returnVideo(String): return a video to the store;
   4. receiveRating(String, int) : take a user's rating for a video; and 5. listInventory(): list the whole inventory of videos in the store.

2. **Objective:** Create a VideoStoreLauncher class with a main() method which will test the functionality of your other two classes. It should allow the following.

   1. Add 3 videos: "The Matrix", "Godfather II", "Star Wars Episode IV: A New Hope".
   2. Give several ratings to each video.
   3. Rent each video out once and return it.

   List the inventory after "Godfather II" has been rented out.

3. Implementation/Code:

```
class Video {
   private String title;
   private boolean checkedOut;
   private double avgRating;
   private int ratingCount;

   public Video(String title) {
      this.title = title;
   }

   public void checkOut() {
      System.out.println(!checkedOut ? (checkedOut = true, "Checked out: " + title) : title + " is already
checked out.");
   }

   public void returnVideo() {
      System.out.println(checkedOut ? (checkedOut = false, "Returned: " + title) : title + " was not checked
out.");
   }

   public void rate(int rating) {
      avgRating = rating >= 1 && rating <= 5 ? ((avgRating * ratingCount + rating) / ++ratingCount) :
avgRating;
   }

   public String details() {
      return "Title: " + title + ", Checked Out: " + checkedOut + ", Rating: " + avgRating;
   }
}

class VideoStore {
   private Video[] videos = new Video[10];
   private int count;

   public void add(String title) {
      videos[count++] = new Video(title);
   }

   public void checkOut(String title) {
      find(title).checkOut();
   }

   public void returnVideo(String title) {
      find(title).returnVideo();
   }

   public void rate(String title, int rating) {
      find(title).rate(rating);
   }
```

```
    public void list() {
        for (int i = 0; i < count; i++) System.out.println(videos[i].details());
    }

    private Video find(String title) {
        for (int i = 0; i < count; i++) if (videos[i].title.equals(title)) return videos[i];
        return null;
    }
}

public class VideoStoreLauncher {
    public static void main(String[] args) {
        VideoStore store = new VideoStore();
        store.add("The Matrix");
        store.add("Godfather II");
        store.add("Star Wars");
        store.rate("The Matrix", 5);
        store.rate("Godfather II", 4);
        store.rate("Star Wars", 5);
        store.checkOut("Godfather II");
        store.returnVideo("Godfather II");
        store.list();
    }
}
```

4. Output:

```
Added video: The Matrix
Added video: Godfather II
Added video: Star Wars Episode IV: A New Hope
Received rating of 5 for video "The Matrix".
Received rating of 4 for video "Godfather II".
Received rating of 5 for video "Star Wars Episode IV: A New Hope".
Video "Godfather II" has been checked out.
Video "Godfather II" has been returned.

Inventory:
Title: The Matrix, Checked Out: false, Average Rating: 5.0
Title: Godfather II, Checked Out: false, Average Rating: 4.0
Title: Star Wars Episode IV: A New Hope, Checked Out: false, Average Rating: 5.0


...Program finished with exit code 0
Press ENTER to exit console.
```

5. Learning Outcomes:

- Designed a functional system to manage video rentals, demonstrating the use of classes and objects in Java.
- Implemented methods for operations like adding videos, renting out, returning, and recording user ratings.
- Applied arrays to store and efficiently manage the video inventory within the store.
- Learned to integrate multiple classes and enable seamless interaction among them in a structured program.